Smart Swimming Pool Monitoring and Control System

Project Description

The **Smart Swimming Pool Monitoring and Control System** is an advanced embedded IoT solution designed to continuously monitor and automatically maintain optimal swimming pool water quality. The system integrates an **STM32 microcontroller** for real-time control and an **ESP32 module** for wireless communication, remote monitoring, and data management.

Multiple sensors are used to measure critical water quality parameters, including **pH level, chlorine concentration, and water temperature**. The collected sensor data is processed in real time using embedded algorithms to determine whether the water conditions remain within safe operating limits.

When abnormal conditions are detected, the system automatically activates **stepper motors and DC motors** to regulate chemical dosing and water circulation, restoring the pool to the desired state. Users can also monitor the pool's status remotely through Wi-Fi connectivity provided by the ESP32, enabling efficient maintenance and reducing manual intervention.

This project demonstrates the integration of embedded systems, sensor interfacing, motor control, wireless communication, automation, and IoT technologies to create a reliable and intelligent pool management solution.

Hardware Components

Component	Quantity	Purpose
STM32 Microcontroller	1	Main controller for sensor processing and automation
ESP32 Module	1	Wi-Fi communication and remote monitoring
pH Sensor	1	Measures the acidity and alkalinity of pool water
Chlorine Sensor	1	Monitors chlorine concentration
Water Temperature Sensor	1	Measures water temperature
Stepper Motor	As Required	Controls chemical dosing mechanisms
DC Motor	As Required	Operates pumps or circulation systems
Motor Driver Module	As Required	Drives stepper and DC motors
CR2032 Battery	1	RTC or backup power supply

Component	Quantity	Purpose
Capacitors (100 nF, 1 µF, 4.7 µF)	Multiple	Power filtering and decoupling
Crystal Oscillator	1	System clock for the STM32
PCB	1	Custom-designed circuit board
Power Supply	1	Provides regulated system power

Key Features

- Continuous real-time monitoring of pool water quality
- Accurate pH, chlorine, and temperature measurement
- Automatic chemical dosing and water regulation
- STM32-based embedded control system
- ESP32 Wi-Fi connectivity for remote monitoring
- Real-time sensor data processing
- Automatic motor and actuator control
- Custom PCB design for compact and reliable operation
- Low-power embedded system architecture
- Safe and efficient swimming pool maintenance

Applications

- Residential swimming pools
- Commercial swimming pools
- Hotels and resorts
- Water parks
- Sports and recreational facilities
- Automated water treatment systems
- Smart IoT-based environmental monitoring